Predicting Post Tommy John Performance.

Final Project Paper

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Introduction.

Major League Baseball has a pitcher health problem. More specifically, a Tommy John surgery problem. The modern game has required pitchers the throw max effort all the time and more often. In 2023 season 35.3% of active Major League pitchers had undergone Tommy John surgery at some point in their career. That is a 29% increase from the 2016 season (American Medical Association). With this increase in Tommy John surgery the question for many front offices has pivoted from how can we keep pitcher healthy to, how can we know this pitcher will be both effective and durable upon return from Tommy John surgery.

This project seeks to shine some light on these problems. First, to predict a pitcher's performance or effectiveness upon returning from Tommy John. Second, will that pitcher remain durable and throw enough innings post Tommy John to be worth maintaining on a Major League roster. MLB front offices are interested in this information because all baseball executives are ultimately graded on one thing winning. Finding effective pitching is a key component to that equation. Additionally, if a front office has the ability to identify which players are more likely to return from surgery effective pitchers that serves as a competitive advantage to that club. This club can then use that information in contract selections and negotiations to both operate more fiscally efficiently and obtain better value talent than its competitors.

This is ultimately a question that can best be answered with data. From the historical records of pitchers who have undergone Tommy John Surgery we can train machine learning models to predict a player's performance post Tommy John. The Dataset used to train these models includes performance metric data from every pitcher in MLB from 2019 to present from Statcast, MLBs' in house database. In all this data set includes 71 different features pertaining to both pitch mechanics and performance. Additionally, Tommy John surgery data was collected from “@MLBPlayerAnalysis”. This dataset includes 42 different surgery and demographic features from all MLB drafted players who have undergone Tommy John Surgery in the modern era.

Methods

Model Introduction

For this project we chose to build two distinct predictive models. The goal of these two models was first to predict a pitchers performance level after returning from Tommy John surgery. This model answers the first question of will a pitcher still be effective after injury. This model was built as a regression model seeking to predict XWOBA or expected weighted on base average against for a pitcher returning from injury. XWOBA is the gold standard for how hitters are evaluated therefore predicting which pitchers are projected to me most effective at suppressing opponent XWOBA will give front offices a tangible evaluation metric to make contract decisions on.

The second model being built seeks to predict pitcher durability. The model built here is a classification model that will predict whether a pitcher will throw 500 pitches in their return season from Tommy John. The MLB median value for pitches thrown in a season is 504. This model is useful for front offices because it answers the question of will this pitcher be durable upon return. If a pitcher is effective at suppressing XWOBA but can only throw 50 pitches they are likely not a smart investment for a front office. MLB contracts are guaranteed money meaning that players are still owed money even if they become injured. Predicting pitcher durability post Tommy John is a very valuable model to MLB front offices seeking to be efficient with their resources.

Data Preparation

To get our data from two sources into one dataset the data was first merged on player name to get one dataset. However, for the evaluation of player performance pre and post Tommy John the data must be summarized by player for one line or record in the dataframe. To achieve this the dataframe was separated into player performances pre and post Tommy John. Both datasets were then summarized by the player down to one record. Each feature was assigned an appropriate aggregate function, mean mode etc. The datasets were then merged back together. To maintain a realistic model, build only the target variables of post Tommy John XWOBA and number of pitches were included from the post-surgery performance dataset.

Exploratory Data Analysis.

Upon exploring our dataset, we wanted to gain an understanding of the differences in target variables between the two classes of pitchers, before and after Tommy John surgery. First, we explored Tommy John surgery's effect on the average number of pitches thrown post-surgery. As seen in figure 1 below it appears that pitchers throw less pitches after Tommy John than before. However, there is no statistical difference between the means of these two groups (p-value: 0.56), suggesting there is no statistical difference in the number of pitches thrown before or after Tommy John surgery.

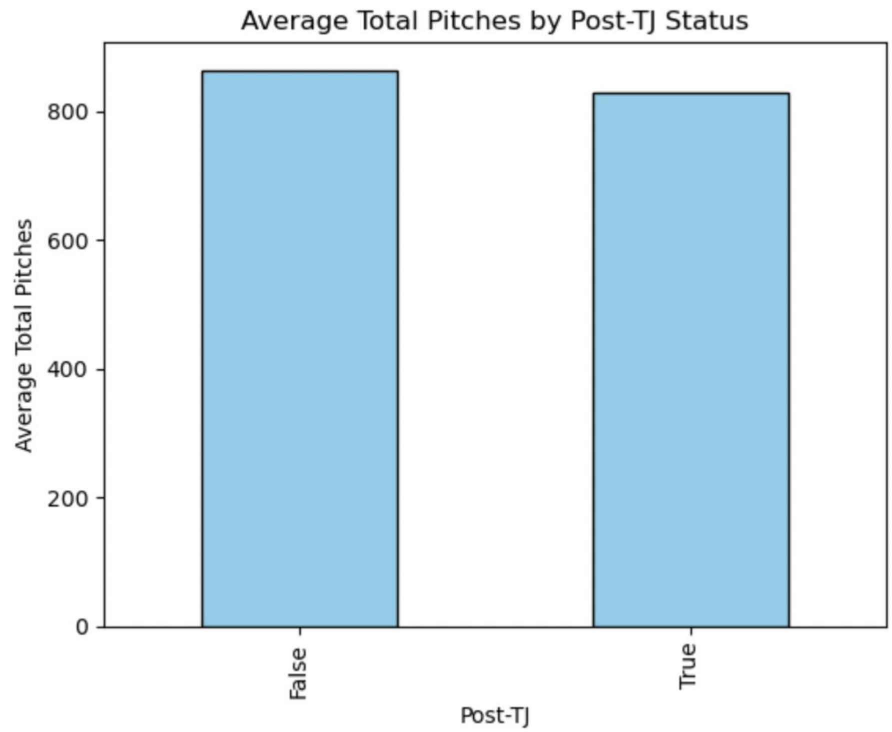

Figure 1: Bar Graph displaying average pitches thrown in a season before and after Tommy John Surgery.

Secondly, we evaluated the difference in XWOBA for pitchers before and after Tommy John surgery. As seen in figure 2 below there is a slight uptick in XWOBA after Tommy John suggesting that pitchers become less effective on average following Tommy John surgery. The difference between average XWOBA before and after Tommy John is statistically different (p-value= 0.01). This information is helpful to front offices knowing that pitchers are likely to perform worse after surgery. This further adds to the value of our predictive model predicting XWOBA post-surgery as finding effective pitchers is likely more difficult than non-effective pitchers as suggested by the bar graph in figure 2.

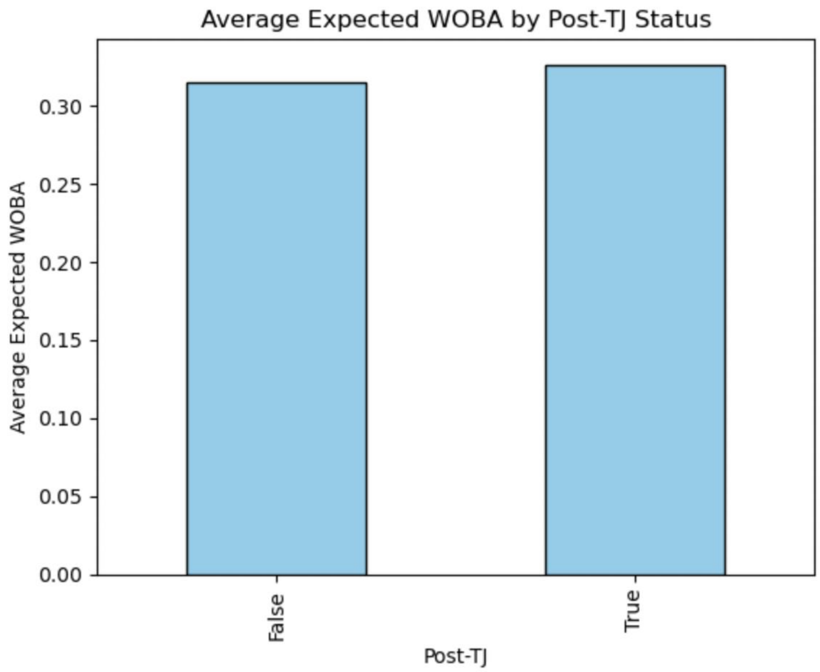

Figure 2: Bar Graph displaying Expected Weighted on Base Average before and after Tommy John Surgery.

Models Trained and Metrics.

For the regression model predicting XWOBA three models were trained. First was a stepwise backwards regression model with a p-value threshold of 0.05 including all features available. The second was a backwards stepwise regression model with the same p-value threshold but not including any of the demographic player information (high school, surgery, etc.). The final regression model trained was a recursive feature elimination model without demographic features as well. The three models were evaluated on goodness of fit (R squared) and root mean squared error. These metrics were selected to both address the error of the predictions as well as the model's ability to explain the variation in the data.

Of the three models trained the recursive feature elimination model performed the best. The backwards regression model with demographic information showed extreme signs of overfitting. Including a training R squared of 1.0. This overfitting was likely due to the model memorizing the training data which included potentially identifying information such as the players high school or surgeon use. The recursive feature elimination model performed the best in test r squared value as well and remained competitive in terms of RMSE. The features selected in this model and evaluation metrics can be seen below in figures 3 and 4 respectively.

```
Recursive Feature Elimination No Dummies
Training R²: 0.31060471199784656, Training RMSE: 0.04387407546154057
Test R²: -0.16992362960771001, Test RMSE: 0.04626991223683221
```

Figure 3: Evaluation metrics for recursive feature elimination model without demographic data.

```
Coefficients and corresponding feature names:
ba: 7.2499
iso: 10.4884
babip: -0.3885
slg: -6.8033
xwoba: -0.2945
xba: 3.5951
velocity: -0.7391
effective_speed: 0.7393
eff_min_vel: -0.7524
obp: 0.3708
xobp: -0.0708
xslg: -3.7440
xbadiff: 3.6548
xobpdiff: 0.4417
xslgdiff: -3.0593
wobadiff: -1.8432
Model intercept:  0.3582879853402157
```

Figure 4: Selected features and coefficients recursive feature elimination model.

The classification model was selected using a grid search cross validation with 5 folds of cross validation. This method balances the bias variation tradeoff and minimizes the effects of overfitting. Five different models were considered in the building of the classification model: logistic regression, random forest, decision tree, bagging (bootstrap aggregation), boosting and gradient boosting. The best model as suggested through the grid search cross validation was the gradient boosting model. This model is evaluated on accuracy. This metric was selected because front offices will use this model to determine if a player is going to be durable or not. The accuracy metric will allow front offices to know how confident they can be in those results. The gradient boosting models evaluation metrics can be seen below in figure 5. Additionally, the feature importance for this classification model can be seen in figure 6. It should also be acknowledged that this model may lean towards being an aggressive acquisition model. As seen in the confusion matrix: figure 7 the model tends to produce false positives. This makes the model more likely to predict a pitcher will be durable where in reality they will not be.

```
Accuracy of the best model and hyperparameters found in the grid search Training Data: 1.00
Accuracy of the best model and hyperparameters found in the grid search Test Data: 0.63
```

Figure 5: Classification model evaluation metrics.

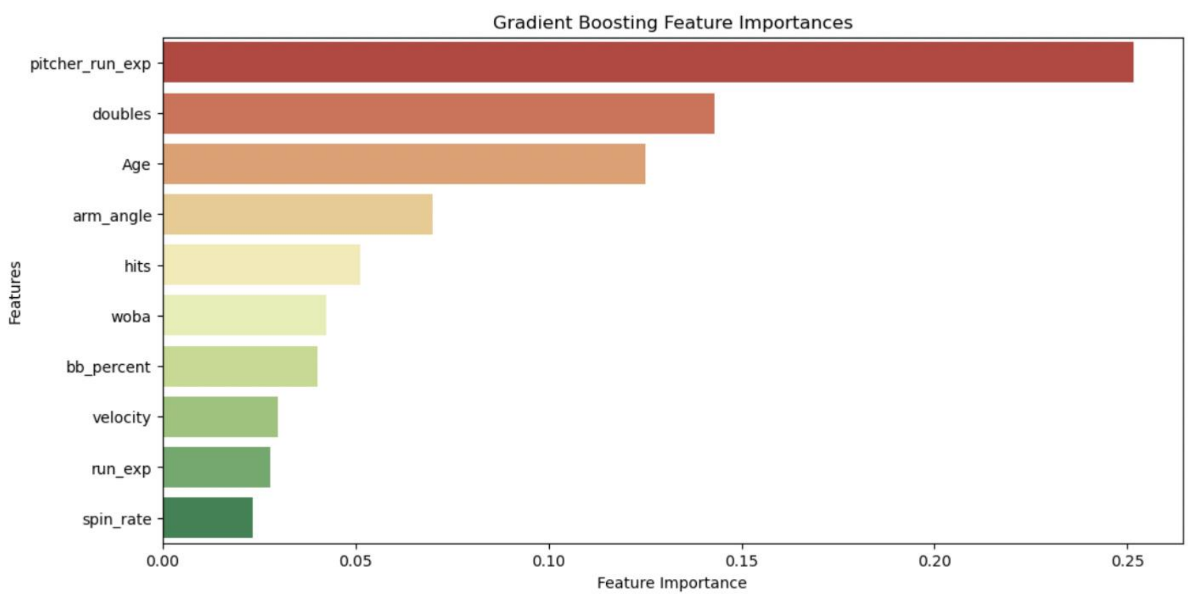

Figure 6: Feature importance pitcher durability classification model.

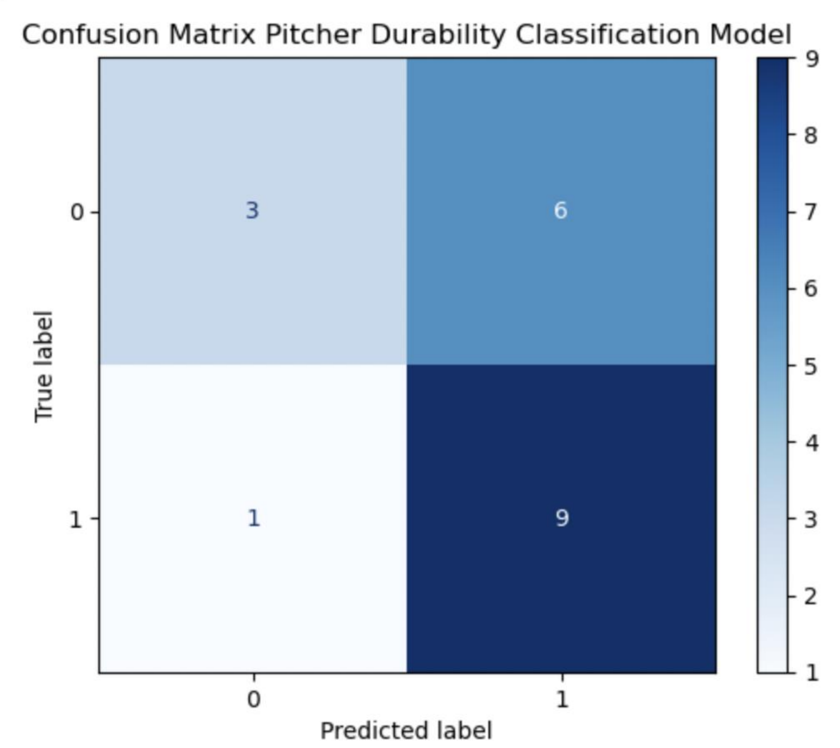

Figure 7: Confusion matrix of the pitcher durability classification model. The label 1 or positive indicates pitching over 500 pitches.

Conclusion

Recursive Feature Elimination Predicting XWOBA

In the end both the model for predicting XWOBA and classifying pitcher durability post Tommy John surgery show promise but suffer from signs of overfitting. In the recursive feature elimination model, we see that the training set far outperforms the test set in both R squared and RMSE. In fact, the model explains none of the variability in the test set as seen by the negative R squared value in figure 3. We did, however, gain some insight on which features drive post Tommy John XWOBA. Although this model may not be ready for full deployment it still has some value for the front office. This model combined with more traditional methods of player selection is sure to give some level of advantage in selecting players with a better likelihood to suppress opponent XWOBA. Additionally, Front offices can look at the selected features and their corresponding coefficients to see how they affect XWOBA. In the fast-paced world of player selection in MLB these informed decisions using the XWOBA regression model could be an advantage over the competition.

To improve this recursive feature elimination model and get it ready for greater influence in contract selections. I recommend that the model is retrained on a larger dataset. Our dataset was restricted to MLB players who have undergone and recovered from Tommy John surgery in the most modern era of baseball (post 2019) this narrows our dataset to only 91 records of pitchers. If the model was expanded to include minor-leaguers and amateur players, it may perform better and see less signs of overfitting.

Gradient Boosting Classification for Pitcher Durability.

The gradient boosting classification model also suffers from overfitting like the model above. This again is likely due to the limited dataset available due to the true scarcity of MLB talent. This model, however, could be deployed right now to some extent. The value of this model is that it can predict with 63% accuracy if a pitcher will throw more than 500 pitches in a season returning from Tommy John. This model should be part of the conversation in MLB front offices when it comes to discussions of if a pitcher's contract should be selected or not. For example, if a pitcher is added in as value in a trade baseball executives should use this predictive classification model to gain perspective on their predicted durability. While making these decisions however, front offices should also take in the models bias to produce false positives or predict durability where pitchers are not durable.

Ethical Implications.

This dataset contains information that is publicly available and easily accessible from MLB as well as other popular media outlets like “@MLBPlayerAnalysis”. The data, however, does include demographic information such as the players' high school or college team. This means that there is potential for the model to make recommendations based off where a player is from, creating a circumstance where certain communities are discriminated against by front offices because the model suggests poor performance. In our models none of these features were selected by either the regression or classification model. If, however, the models we retrained on a larger dataset this may not be the case. Additionally, this model has the potential to affect pitchers' career or livelihoods if implemented. If the model predicts poor performance or poor durability many former major leaguers may be out of work with no other employable skills beyond baseball.

Resources

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